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The Editor-in-Chief
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Dear Editor,

I submit the Original Article **"Retinal foundation-model versus ImageNet pre-training for diabetic retinopathy grading across datasets: how much of the external performance drop reflects the reference standard?"** for consideration in *International Ophthalmology*.

Automated diabetic retinopathy (DR) grading performs well on the data it is developed on but loses accuracy when the camera, population or grading protocol changes. Retinal foundation models such as RETFound are often expected to solve this problem. This study tests that expectation directly. It also asks a question that matters to every ophthalmologist who reads an AI validation paper: how much of an "external performance drop" is model failure, and how much reflects a different case mix or a different reference standard?

Models were developed on APTOS 2019 and evaluated once, unchanged, on Messidor-2 (adjudicated specialist grades) and the official IDRiD release. The pre-specified primary analysis, with multiplicity control, found that fully fine-tuned RETFound degraded neither less nor detectably more than an ImageNet-pretrained EfficientNet-B3. The study then shows that:

- most of the Messidor-2 drop disappears once case mix is matched and no/mild DR are merged, and the two external datasets fail in opposite directions at that boundary, which points to differences in the reference standard rather than to model fragility;
- RETFound's apparent calibration advantage is reproduced by label smoothing alone, a training-recipe effect rather than a benefit of retinal pretraining;
- confidence-based referral halved missed referable DR on one external dataset but not the other, and simple confidence was as informative as more elaborate uncertainty scores;
- tested quantitatively against expert lesion masks, saliency maps localised lesions only weakly, so they should not be read as proof of interpretability.

The study is therefore an evaluation of cross-dataset reliability, not a proposal for another classifier. It is reported following TRIPOD+AI. It includes an audit that found 5.9% of labels in a widely shared IDRiD mirror differed from the official release, and it provides per-image predictions for every model. The work adds to recent articles in this journal on deep learning for DR and retinal imaging by emphasising external testing, confidence intervals and quantitative evaluation of uncertainty and explainability.

I confirm that this manuscript is original, has not been published previously, and is not under consideration by any other journal. It has not been posted as a preprint. I am the sole author and have approved the submitted version. The study used only public, de-identified datasets. It received no funding, and I have no competing interests. The use of AI assistants in preparing the code and text is described in the Methods. I would prefer the standard (subscription) publication route.

Thank you for considering this submission.

Yours sincerely,

Pravin Raj E